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Paper

Short-Term Building Load Forecast Based on a Data-Mining Feature Selection and LSTM-RNN Method

Gaiping Sun*[a], Non-member
Chuanwen Jiang*, Non-member
Xu Wang*, Non-member
Xiu Yang**, Non-member

Short-term load forecast for individual electric customers is becoming increasingly important in the grid operation, since the power system is becoming a more interactive and intelligent system. Accurate short-term load forecast for industrial or commercial electric buildings is more challenging due to the complicated load characteristics and numerous influence variables. In this paper, we consider maximizing the relevancy and minimizing the redundancy criterion to select effectively feature variables, which influence the building load consumption, and then a deep learning technique—long-short memory recurrent neural network is proposed to predict the load consumption. This novel strategy captures distinct load characteristics, choosing accurate input variables, and shows a great forecasting performance as demonstrated by three different types of city building load in China. © 2020 Institute of Electrical Engineers of Japan. Published by John Wiley & Sons, Inc.

Keywords: short-term load forecast; influence factors; feature selection; LSTM-RNN

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1. Introduction

Short-term load forecast means forecasting load consumption from half an hour to 1 week ahead. Accurate short-term load forecast is a compulsory work for modern power system operation [1]. With the increasing penetration of renewable generations, different kinds of active load, energy storage system, the distribution system is becoming more complex and uncertain. Accurate short-term load prediction for the electric consumers can help the operator to prevent the potential substation overloading, assist demand side management to shift reasonable load, balance the generation and consumption, and reduce gases emissions [2].

The installation and use of smart meters have collected amount of energy consumption data, which make the forecast at the individual customer possible. However, Forecast for the individual customers (individual building level, such as hospital, school, market, and hotel, etc.) is more difficult because of the complicated energy consuming behavior and the uncertainty influencing factors [3]. The distinct load characteristics could make electric load curve shape not as smooth or continuous as a large area load curve shape. The irregular and unstable distribution load increases the difficulty of load forecast.

In the past decades, many approaches have been proposed to predict short-term load consumption; the methods are mainly divided into two categories: classical statistical methods and artificial intelligent algorithms. The former methods include parametric regression models such as polynomial functions, autoregressive moving average, multiple linear regression [4,5], etc. Regression models [6] aim to find the relationships between load target and affecting variables (e.g. climate index, weekday index, and weekend index). Kalman filter is also an attractive method in view of its recursive property, in a Kalman filter forecasting model [7], load was developed in the state space formulation including linear state and measurement equations. However, those models are mostly linear components though the load series are basically nonlinear relationships of the exogenous variables. Artificial intelligent algorithms which are developed from the late 1980's have been widely applied in load predicting problem, such as fuzzy logic system, expert system, etc. Artificial neural networks [8,9] and support vector machines [10] have attracted a great attention and applied extensively in view of its stronger study ability to map the nonlinear relationship between the input variables and output targets compared with other approaches. Another machine learning technique, random forest [11] is also proposed for load forecast. In Ref. [12] a similar load day-based wavelet neural network is introduced to predict load demand. Similar days identified by correlation analysis are considered as the inputs of neural network, wavelet decompose load series to low and high frequencies, then each separate neural network was used to predict future load. Recently, combination of linear method and intelligent algorithms are applied to enhance the prediction accuracy, in Ref. [13] a hybrid forecasting method consists of a forecast-aided state estimator and a multilayer perceptron (MLP) neural network is presented and the result illustrates a better predicting performance. In Ref. [14] a fuzzy system that combines with an evolutionary algorithm is proposed to perform an efficiently short term load forecasting result.

Except predicting algorithm, the input selection of predicting methods cannot be ignored. Researchers always choose input variables by experience or simplified criterion [15,16]. In order to obtain all the potential influence variables, [13] considers amount of input variables, which contains load demand of several weeks ahead and climate parameters of several weeks ahead. It increases the forecasting model complexity, prolong training period, and even make redundant information. Similar day selection is another

[a] Correspondence to: Gaiping Sun. E-mail: gaiping_sun@sjtu.edu.cn

*School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai, 200240, China

**School of Electric Power Engineering, Shanghai University of Electric Power, Shanghai, 200090, China

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method to improve forecasting performance. The similar day's load is selected as the input load of a forecasts model. In Ref. [12], the criterion to select similar days is based on the minimum sum of absolute errors of the wind chill temperature between the history day and forecasted day.

All of the above load forecasting methods are mainly discussed at the substation or system level. With the development of smart grid, load forecast for the individual users (residential, industry, buildings, etc.) are gaining enhancing interest. Stephen *et al.* [17] clustered daily historical load data of individual households. Those households were considered as label sequences. Then a Markov chain-based sampling technique derived from practice theory of human behavior was adopted to forecast day ahead load. Ref. [18] used a cluster method and decision tree to select similar load forecasting method for load forecast. Deep learning is a high efficiency machine learning technology in the field of computer science. The computational models of deep learning are composed by multiple processing layers, which learn representations of data with multiple levels of abstraction. These methods have made remarkable achievements at different fields in recent time, such as speech recognition, visual object recognition, object detection, and many other domains such as drug discovery and genomics [19]. Time series forecasting research also have been attempted to adopt deep learning method [20]. A pooling-based deep recurrent neural network (RNN) was proposed in Ref. [21], the forecasting method batched a group of customers' load profiles into a pool of inputs, and the result is tested on 920 smart-metered customers from Ireland, which shows an improved forecasting accuracy compared with other methods. Weicong Kong *et al.* [22] clustered the residential load and then built a long short-term memory (LSTM) RNN-based framework to predict short-term load. As a result, the framework outperforms the other algorithms for individual residential households. Ref. [23] constructed a deep learning neural network to forecast nonresidential customer load, and the result illustrated an effective forecasting accuracy. Therefore, deep learning method shows a great ability in distinguishing the uncertainty pattern for the distribution consumer load.

There had been some research studies on residential load forecast, while there was few study on small-scale buildings load forecast. For an urban commercial building, the electricity equipment constitution is distinct. In general, weather is the most important influence variables of electric consumption. For example, air conditioning equipment consume huge amount of electric energy at tropical or chilly climate days, while electric energy consumption of lighting equipment depends on sunshine degree. The main object of electric consumption is to keep people's comfort, which can be defined as human body amenity. Temperature is always analyzed as the main weather variable in a load forecasting model, while other relative variables are often discussed in the literature, such as humidity, wind speed, sunshine, and air quality are also play a vital role in human body comfortableness during electric load consumption.

This paper introduces a novel short-term load predicting framework which concludes a comprehensive input variable selection process and a LSTM network forecasting model. The main contributions are provided as follows:

1. The climate impact factors are mined deeply by considering human body comfortableness of urban buildings. Except the normal climate variable such as temperature, humidity, other variables, for instance, heat index (HI), and air quality are also considered as the influence variables.

2. The best candidate input features for different types of urban building are selected by a max relevancy and min redundancy (MRMR) criterion based on Pearson's correlation coefficient. The selection of input features is helpful for the forecasting accuracy and efficiency.

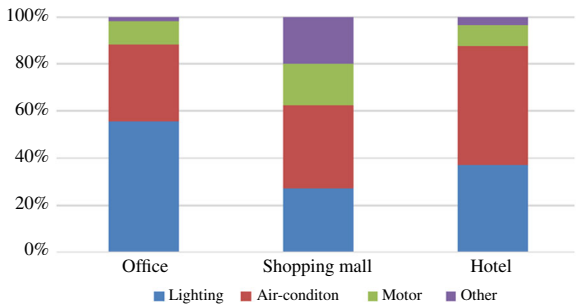


Fig. 1. Load composition of three different types of buildings in a summer day

3. A deep learning method—a LSTM-RNN for building load forecast is presented in this work. This kind of forecasting model preforms a great predicting capacity compared with other models based on the same data sets.

The remaining sections are organized as follows. In Section 2, the input selection process is represented; it concluded the influence factors analysis and the feature selection process. In Section 3 the forecast strategy, a LSTM-RNN is introduced in details. Then the application of the mentioned strategy and numerical result for three different types of city buildings load are presented and discussed in Section 4. A brief conclusion of this paper is summarized in Section 5.

2. Proposed Feature Selection

2.1. Influence factors analysis Daily load demand on the consumer side has a complicated characteristic which depends on a series of qualitative and quantitative variables. Qualitative variables consist of weekday index, weekend index, holiday index, and seasonal index. While quantitative variables mainly refer to meteorological factor like temperature, humidity, sunshine, wind speed, etc. [24].

For city small-scale building load, climate-sensitive load demand takes over a significant portion of the total load demand in order to keep a comfortable body environment. Figure 1 shows the load compositions of three different commercial buildings in China Shanghai in a summer day. It is obvious that air-conditioning and lighting systems consume most of the electricity energy. Temperature is always analyzed as the main weather variable in a load forecasting model, while other relative variables are often discussed in the literature. Humidity and wind speed also play a vital role in human body comfortableness during the hot months or cold months. Human body amenity is a comprehensive and effective biological meteorological evaluation in accordance with human's reaction of different climate [25,26]. The expressions of human body comfort index are distinct because of the differences in geographical environment and personal feelings. National Oceanic and Atmospheric Administration explains HI, which represents human-perceived equivalent temperature. It is a measure of how hot it really feels when other climax variables (relative humidity, wind speed, etc.) are factored in with the actual air temperature [27]. HI takes temperature and humidity into consideration at a shady area and the formulas are constructed as:

$$\begin{aligned} \mathrm{HI}_t = & -42.379 + 2.04901523 \times T_t + 10.14333127 \\ & \times \mathrm{RH}_t - 0.22475541 \times T_t \times \mathrm{RH}_t - 6.83783 \times 10^{-3} \\ & \times {T_t}^2 - 5.481717 \times 10^{-2} \times {\mathrm{RH}_t}^2 + 1.22874 \times 10^{-3} \\ & \times {T_t}^2 \times \mathrm{RH}_t + 8.5282 \times 10^{-4} \times T_t \times {\mathrm{RH}_t}^2 - 1.99 \\ & \times 10^{-6} \times {T_t}^2 \times {\mathrm{RH}_t}^2 \end{aligned} \quad (1)$$

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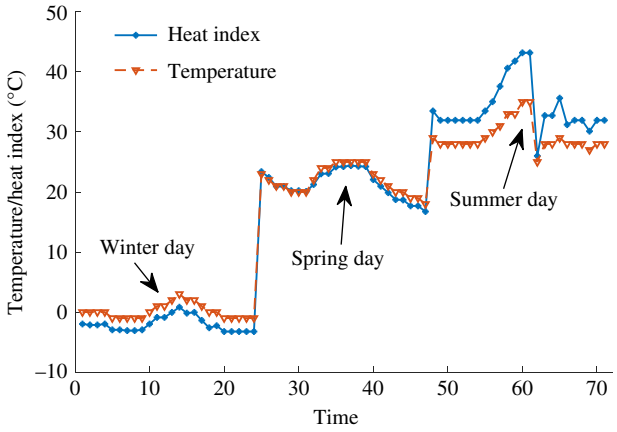


Fig. 2. Heat index via actual temperature in 2012

$$\mathrm{HI}_t = 0.5 \times [T_t + 61.0 + (T_t - 68.0) \times 1.2 + \mathrm{RH}_t \times 0.094] \quad (2)$$

where T_t is given in degrees Fahrenheit (°F) at time t. RH_t represents relative humidity at time t. When $80\,^\circ\mathrm{F} \le T_t \le 112\,^\circ\mathrm{F}$, the first formula is applied to calculate the HI, while $T_t < 80\,^\circ\mathrm{F}$, the second formula is used as the calculated HI.

The HI via actual temperature value of 3 days (a winter day, a spring day and a summer day) is provided in Fig. 2 in shanghai, China at 2012. Overall speaking, in hot days, the human-perceived equivalent temperature is higher than actual temperature; in cold days, the human-perceived equivalent temperature is lower than actual temperature. Instead during comfortable days there is little difference between them.

In addition to the normal weather influence factors, the air pollution issue is a serious problem in China in recent years. The air quality affects the lifestyle of urban dwellers, including electricity-consuming behavior. For instance, in a poor air quality day, people maybe change the early going out plan, choose to stay at home, and turn on the air purification equipment. On another hand, when the air pollution is particularly serious, the government may take emergency measures to improve air quality, such as large thermal power plants reduce production, industrial load consumption could be limited. This series of conditions could change the place and manner of electricity consumption, and reduce the load prediction accuracy of conventional prediction method.

Therefore, sufficient and adequate influence factors are considered for the building load forecast; in this paper, temperature, humidity, HI, wind speed, and air pressure are analyzed. Besides, air quality index (AQI), PM2.5 (particles with an aerodynamic diameter of 2.5 μm or less), PM10 (particles with aerodynamic diameter equal to or less than 10 μm), SO_2, and NO_2 are also considered as the influence factors.

2.2. MRMR-based feature selection Few input feature variables cannot provide enough information, while excessive variables may lead to redundancy. Feature selection is always used to select features with a high correlation with the output and a low correlation between themselves. Many methods have been used to reduce the numbers of variables in the data mining problem, such as mutual information, Relief-F, information gain, etc. In this paper, MRMR combined with Pearson's correlation method is used to select the best candidates and exclude redundant features.

Pearson's correlation coefficient is used to calculate the relationship strength between two vectors X and Y it can be presented as follows:

$$\rho(X,Y) = \frac{\mathrm{cov}(X,Y)}{\sqrt{\mathrm{var}(X) \times \mathrm{var}(Y)}} \quad (3)$$

$\mathrm{cov}(X, Y)$ represents covariance of X and Y, $\mathrm{var}(X)$ and $\mathrm{var}(Y)$ represent the standard deviation of X and Y. When X and Y are point samples with a total number of K, $X = \{x_i, i = 1, 2, \ldots, K\}$, $Y = \{y_i, i = 1, 2, \ldots, K\}$, then the sample correlation coefficient can be expressed as:

$$\rho(x,y) = \frac{\sum_{i=1}^{K}(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum_{i=1}^{K}(x_i - \overline{x})^2}\sqrt{\sum_{i=1}^{K}(y_i - \overline{y})^2}} \quad (4)$$

where $\overline{x}$ and $\overline{y}$ are the average value of X and Y.

Regarding the feature selection problem, X is defined as the input variable vectors with m features (such as temperature, humidity, wind speed, etc.) and y is the target vectors (load consumption). The purpose is to find set X_n with n features ($n < m$), which jointly have the largest dependency on the target vectors. This scheme is named Max-Dependency; it can be represented by the following form:

$$\begin{cases} \max D(X,y) \\ D = \frac{1}{n}\sum_{i=1}^{n}\rho(x_i,y) \end{cases} \quad (5)$$

$\rho(x_i, y)$ is the Pearson's correlation value between input features and output values. Then min-redundancy scheme is added to select mutually exclusive features:

$$\begin{cases} \min R(X) \\ R = \frac{1}{{C_n}^2}\sum_{i=1}^{n-1}\sum_{j=i+1}^{n}\rho(x_i,x_j) \end{cases} \quad (6)$$

$\rho(x_i, x_j)$ is the Pearson's correlation value between input vectors. ${C_n}^2$ is the total combination number of different input variables.

The criterion which combines the above two constraints is named maximal relevance minimal redundancy. The operator is defined as follows,

$$\begin{cases} \max \phi(D,R) \\ \phi(D,R) = D - R \end{cases} \quad (7)$$

where D and R are result calculated by (5) and (6).

Therefore, in this solving process, we have selected the candidate subsets of X_{k-1}, with $\{k-1\}$ features. Then, the kth feature from feature subsets is $\{X - X_{k-1}\}$. So, X_k can be calculated as:

$$\max_{x_i \in X - X_{n-1}} \left(\rho(x_i,y) - \frac{1}{n-1}\sum_{j=1}^{n-1}\rho(x_i,x_j) \right) \quad (8)$$

The algorithm flow is presented in the following,

Input: input training dataset $X \in R^{n \times m}$, output variable Y, full feature set S, target feature size S_s.

Calculate Pearson's correlation $\rho(x_i, y)$, select the feature variable $x_{\max}$ with the biggest correlation value, construct target feature subsets S_s, and available feature $S_a = S - S_s$

For $k = 2$ to s

For each feature S_j in S_a and each feature in S_i in S_s, search for the new feature x_k according to : $x_k = \max_{x_i \in S - S_{k-1}} \left(\rho(x_i,y) - \frac{1}{k-1}\sum_{j=1}^{k-1}\rho(x_i,x_j) \right)$

Update the value $S_s = S_s \cup \{x_k\}$, and $S_a = S_a - \{x_k\}$

End

Output S_s

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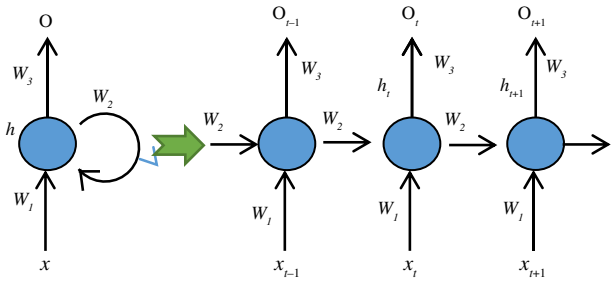


Fig. 3. Structure of recurrent neural network (RNN) and unrolled structure of RNN in time

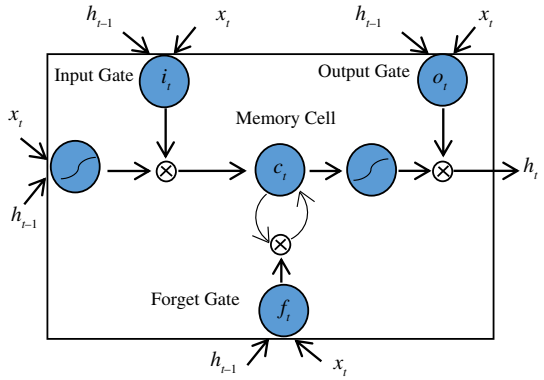


Fig. 4. Structure of a long short-term memory cell

3. Forecast Method

3.1. LSTM-RNN network In traditional neural networks, there are only connections between adjacent layers, no connection among the nodes in the same layer. As a result, all inputs and outputs are independent of each other. It is difficult to deal issues, which have temporal interactions.

RNNs with loops in them address this issue. Figure 3 shows a typical RNN architecture (left) and its unfold in time (right). In this network, the hidden units receive a feedback from the previous state to current state, allows information to be passed from one step to the next step. The unrolled structure can be considered as multiple copies of same network. It can be found that the final output is dependent on both the current input and the output of previous hidden layer.

However, classical RNN have difficulty in long-range dependencies where the gradient either explodes or vanishes during back propagation, which limits the ability to learn long-term temporal correlations [28]. RNN with LSTM cells is designed to avoid the long-term dependency problem, remember information for long periods of time. The original model of LSTM is first introduced by Hochreiter *et al.* in 1997 and improved by Gers *et al.* with an extra forget gate [29]. It is a special type of RNNs capable of learning order dependence, and was refined and popularized by many people on a large variety of problems, including sequence prediction problems.

The unit in a RNN applies an element wise nonlinearity to the transformation of inputs and recurrent units in a simple way. However, this structure in LSTM has an internal recurrence (a self-loop), in addition to the outer recurrence of the RNN, which is shown in Fig. 4. Let $\{x_1, x_2, \ldots, x_T\}$ refer to a typical input time series for a LSTM, x_t represents the feature dataset used to the input at time step t, the LSTM cell is equipped with a memory cell c_t, which stores the long-time dependencies. During the training process, the input gate i_t, output gate O_t, and forget gate f_t allow the LSTM to forget or write new information to the memory cell. The detail formulas of input gate, forget gate, cell state, output gate, and output hidden state are represented in Fig. 4.

$$i_t = g(W_{xi}x_t + W_{hi}h_{t-1} + b_i) \tag{9}$$

$$f_t = g(W_{xf}x_t + W_{hf}h_{t-1} + b_f) \tag{10}$$

$$c_t = f_t c_{t-1} + i_t \tan h(W_{xc}x_t + W_{hc}h_{t-1} + b_c) \tag{11}$$

$$o_t = g(W_{xo}x_t + W_{ho}h_{t-1} + b_o) \tag{12}$$

$$h_t = o_t \tan h(c_t) \tag{13}$$

where the initial hidden state h_0 is set to zero, $g(\cdot)$ is the sigmoid function. i_t, f_t, c_t, and o_t are the input, forget, memory, and output gate at time step t. they are all sigmoid functions that can be zero, indicating the ability to restrict the calculating information to the next node. W represents the weights of each gate in the network, they perform the transformation between two components. For example, W_{xf} means the input-forget gate matrix, W_{hi} represents the hidden-input gate matrix. W_{hc} represents the hidden-memory gate matrix. b is a bias added to formula at each gate, aims to enhance the flexibility to fit the data of the network.

At last a fully connected layer shows a linear transformation on the output hidden state h_t to calculate a load power value at time step t. the formula is provided as:

$$P_t = V_{\text{out}}h_k + b_y \tag{14}$$

where V_{out} is the fully connected layer's weight matrix and b_y is the biases.

The unknown network weights and biases are updated by an optimization method called Adam. The method is based on the gradient of a loss function. The loss function in this paper is defined by

$$E = \sum_{t=1}^{N} \frac{1}{2}(P_t - P_t^{'})^2 \tag{15}$$

where N is the length of the time series P_t, $P_t^{'}$ is the actual output load power. The update process is performed as,

$$s_t = \beta_1 s_{t-1} + (1-\beta_1)\nabla E(W_{t-1}) \tag{16}$$

$$r_t = \beta_2 r_{t-1} + (1-\beta_2)\nabla E^2(W_{t-1}) \tag{17}$$

$$\widetilde{s_t} = s_t/(1-\beta_1^t) \tag{18}$$

$$\widetilde{r_t} = r_t/(1-\beta_2^t) \tag{19}$$

$$\Delta W = -\alpha \frac{\widetilde{s_t}}{\sqrt{\widetilde{r_t}} + \varepsilon} \tag{20}$$

$$W_t = W_{t-1} + \Delta W \tag{21}$$

where β_1 and β_2 are exponential decay rates for the moment estimates, they are set to 0.9 and 0.999. α is the step size set to 0.001. ε is a constant parameter set to 10−8.

3.2. LSTM-based forecasting framework Figure 5 shows the short-term load forecasting framework introduced in this paper. There exist multi-input variables of time series which are selected by the MRMR criterion, such as temperature,

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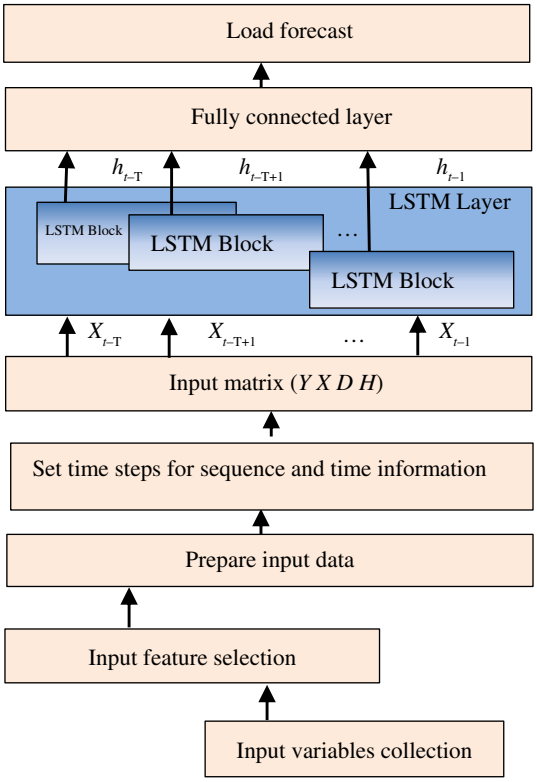


Fig. 5. Proposed forecasting flowchart

air humidity, HI, wind speed, air pressure, and other relative air quality index. Therefore, multi-LSTM neural networks are paralleled to accept different important information in time series. LSTMs learn the distinct load consumption characteristics through different correlation information. In the frameworks, the steps are introduced as follows,

(1) The variable factors that could affect consumers' energy consumption pattern are collected and reprocessed. In this study, the following features are considered,

$$X = \{x_{\text{tem}}, x_{\text{hum}}, x_{\text{hi}}, x_{\text{ws}}, x_{\text{air}}, x_{\text{AQI}}, x_{\text{pm2.5}}, x_{\text{pm10}}, x_{\text{SO}_2}, x_{\text{NO}_2}\}$$

where x_{tem} represents temperature, x_{hum} means air humidity, $x_{\text{hi}}$is the HI calculated by (1) and (2), x_{ws} means wind speed, x_{air} represents air pressure, $x_{\text{AQI}}$is the air quality index, $x_{\text{pm2.5}}$ and x_{pm10} are particles with aerodynamic diameter equal to or less than 2.5/10 μm. $x_{\text{SO}_2}, x_{\text{NO}_2}$ are the values of SO_2 and NO_2 in the air.

(2) Variable features mentioned above are sorted and selected by MRMR criterion with Pearson's values. According to the calculating result of different buildings at four seasons, high-related and low-redundancy variable are chosen as the input sequence variables of LSTM RNNs.

(3) The inputs of the framework are time series data and corresponding time information:

- Load consumption sequence with past T_1 time steps for look-back are built. It contains consecutive T_1 time points before the forecast time point.

$$Y = \{Y_{t-T_1}, Y_{t-T_1+1}, \ldots, Y_{t-1}\} \in \Re^{T_1}$$

- The series of inputs X represent the variables selected by steps (1) and (2). Those time series consist of time points with past T_2 time steps for look back, and the predicted climate and air quality value at the forecasted time. $X = \{X_{t-T_2}, X_{t-T_2+1}, \ldots, X_{t-1}, X_t\} \in \Re^{T_2}$
- Time information contains corresponding day of week indices for the past T time steps D, which ranges from 0 to 6, and corresponding binary holiday marks H, which is defined as 0 or 1.

(4) LSTM-RNN is sensitive to the data scale, input time series data are normalized to (0, 1) with a Min-Max method and time information variables are encoded by one hot encoder. Once all the vectors are normalized, they are concatenated as Multivariable inputs to LSTM blocks. They are fed to the LSTM blocks in the LSTM layer.

Then the output of the LSTM layer is fed to a fully connected layer that integrates all the LSTM layer output. Through the fully connected layer, the target (i.e. load consumption forecast) is obtained.

4. Performance Evaluation

The proposed load demand forecasting strategy is illustrated on three different types of city buildings (an office, a shopping mall and a hotel) in China Shanghai. The load demand data set of the whole year 2017 is obtained from power dispatching department while the weather parameter of that year is downloaded on the weather forecast website. In order to verify the performance of the proposed method with all typical result, 4 weeks of each seasons on year 2017, including January 9–15, May 8–14, August 7–13, November 6–12, are considered as the testing week.

In order to illustrate the forecast performance of the proposed method, except the designed method, some other models, which include an empirical method, and some popular machine learning algorithms are adopted for comparison. All of the frameworks are built on a desktop PC with a 3.2 GHz AMD Ryzen 72 700 Eight-Core processer and 16G memory.

Autoregressive integrated moving average (ARIMA), is a classic linear method always used to be a predictor. ARIMA algorithm considers both the dependency relationship and the random fluctuations between the inputs and outputs. In a standard ARIMA (p, d, q), d is the degree of differencing process, p and q are the autoregressive model order and the moving average model order. According to the analysis of the dataset in this paper, the RIMA model is defined as (1, 1, 0).

Back propagation neural network (BPNN) is a classic artificial neural network that has a strong study ability to find the nonlinear relationship between inputs and output. It has been widely used in the forecasting problem. One hidden layer MLP is illustrated to be a sufficient regression function, which is able to be a universal approximation of nonlinear problems. Levenberg–Marquardt is applied as the iterative optimization algorithm based on calculation of cost function gradient. The hidden nodes in this paper are set to 32.

BPNN model with similar day's selection (BPNN-SD) is an improved method, which chooses temperature-based similar days as the training samples of neural network. The criteria to select similar days are based on the minimum sum of absolute errors of the temperature between the history day and forecasted day in this paper.

Proposed method (MRMR-based RNN-LSTM). In this experiment, Keras library is employed to establish the forecast strategy. The number of nodes in the hidden layer is 50. The activation function in the LSTM layer is relu. Adam is used for the computational optimizer. The number of training epochs is set to 100.

In order to evaluate the performance of the proposed method, root mean square error (RMSE) and mean absolute percentage error (MAPE) are calculated as estimation of the predicting

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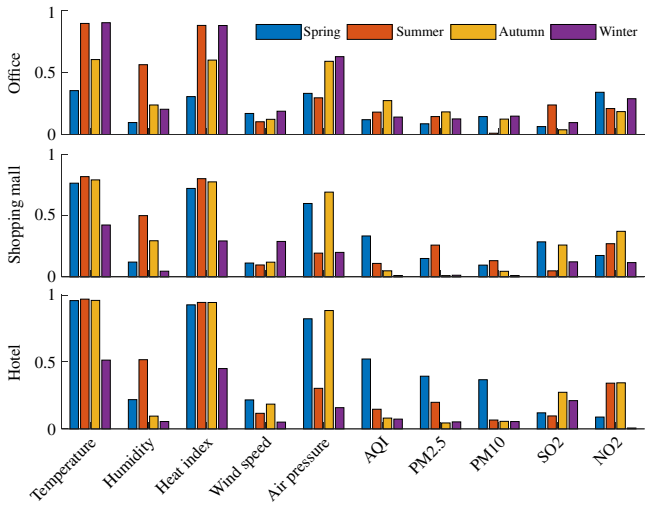


Fig. 6. Pearson's correlation of different city buildings

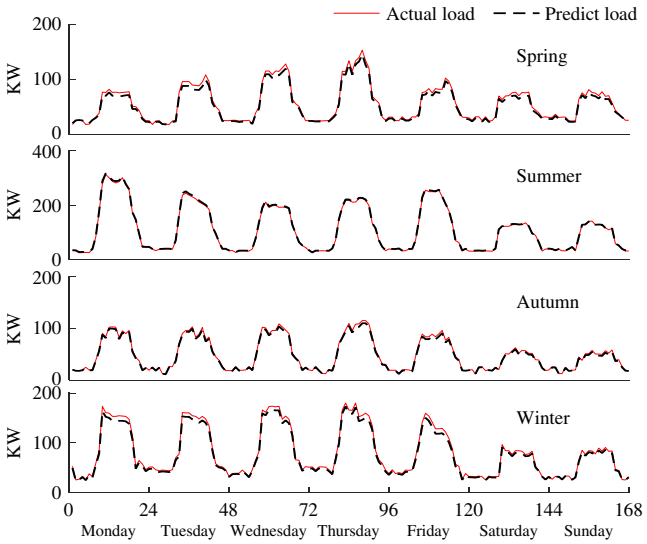


Fig. 7. Weekly forecast result at each seasons in an office building

efficiency. The functions of the two indicators are defined as:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{t=1}^{N}(L^d(t) - L(t))^2} \tag{22}$$

$$\mathrm{MAPE}\ (\%) = \frac{1}{N}\sum_{t=1}^{N}\frac{L^d(t) - L(t)}{L(t)} \times 100 \tag{23}$$

where $L^d(t)$ is the foresting load demand. $L(t)$ is the actual load consumption. N represents the hours of the forecasting horizon. In this paper, one continues week of daily forecasted load demand value are simulated, which means $N = 168$.

4.1. Feature selection result The Pearson's correlation between the mentioned climate factors and electricity load are calculated at different seasons. The result is shown in Fig. 6. For the entire building load, temperature and HI take over the most important influence role. HI is composed by several different influent factors; however, the correlation result shows that it has similar influence degree with temperature. Humidity and air pressure influence the load consumption in some detailed seasons. For example, humidity shows importance in summer for all the buildings, while air pressure cannot be ignored in spring and summer, especially for the shopping mall and hotel buildings. The air quality index does not present a decisive relationship to the load consumption. However, there may be a weak relationship between some air quality index and load consumption, such as NO_2, PM2.5, and PM10. Those index influences people's electricity consumption in some cases.

Then MRMR-based input selection result of different buildings is shown in Table I. Temperature is the most important influence factor in every building at each season. Although HI is calculated based on other variables such as temperature, humidity, etc., it still plays an irreplaceable role in the feature selection process. Humidity is an important factor in summer and spring, while air pressure affects the load consumption mainly in spring and winter. Since the load is consumed indoor, wind speed does not show an important impaction on electricity consumption. Compared with climatic factors, air quality index represents a weak relationship with load consumption. Among them, NO_2 takes part in the middle position by the feature selection. In this paper, the top six features are selected as the input variables of each forecasting model.

4.2. Short-term load forecast result The proposed model is illustrated for one-hour ahead load forecast of one continuous week at each season, in order to test the graphical view of proposed method. Figures 7, 8, and 9 show the forecasting result curve at each season of three different buildings. In those figures, the forecasting load curve and the real load curve are plot. It can be found that the proposed strategy's curve is quite close to the real load data curve. For an office load, the load fluctuation is regular, which has a single peak each day, a high-level load consumption on weekday and low-level load consumption on weekend. For a shopping mall building load, the load shape has two peaks each day; there is not obvious difference between weekdays and

Table I. Max relevancy and min redundancy feature selection result of different buildings

	Spring			Summer			Autumn			Winter		
Variables	Office	Mall	Hotel	Office	Mall	Hotel	Office	Mall	Hotel	Office	Mall	Hotel
Temperature	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Humidity	**3**	**6**	**4**	**3**	**3**	**3**	**4**	**5**	**6**	**5**	**3**	**3**
Heat index	**4**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**
Wind speed	7	8	8	7	7	7	6	7	7	**6**	**5**	7
Air pressure	**6**	**3**	**3**	**6**	**5**	**5**	**3**	**3**	**3**	**3**	7	**6**
AQI	9	7	**5**	8	10	9	8	9	9	9	9	9
PM2.5	10	10	10	10	9	10	10	10	10	10	10	10
PM10	8	9	7	9	8	8	9	8	8	8	8	8
SO_2	**5**	**4**	9	**5**	**6**	**6**	7	**6**	**5**	7	**6**	**4**
NO_2	**2**	**5**	**6**	**4**	**4**	**4**	**5**	**4**	**4**	**4**	**4**	**5**

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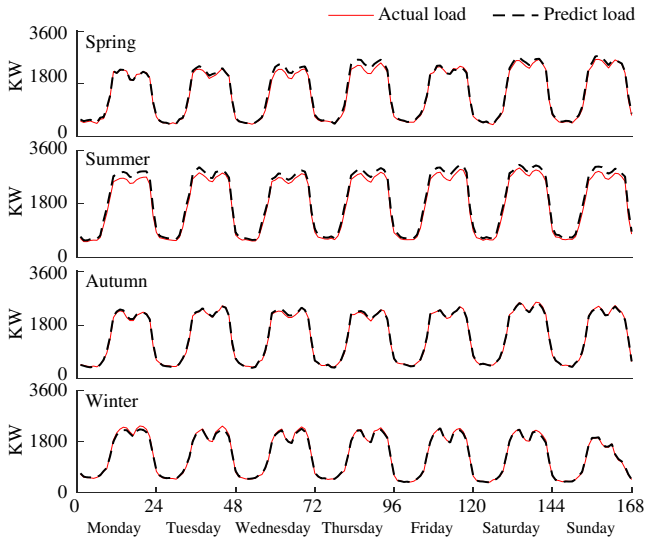


Fig. 8. Weekly load forecast at each season in a shopping mall building

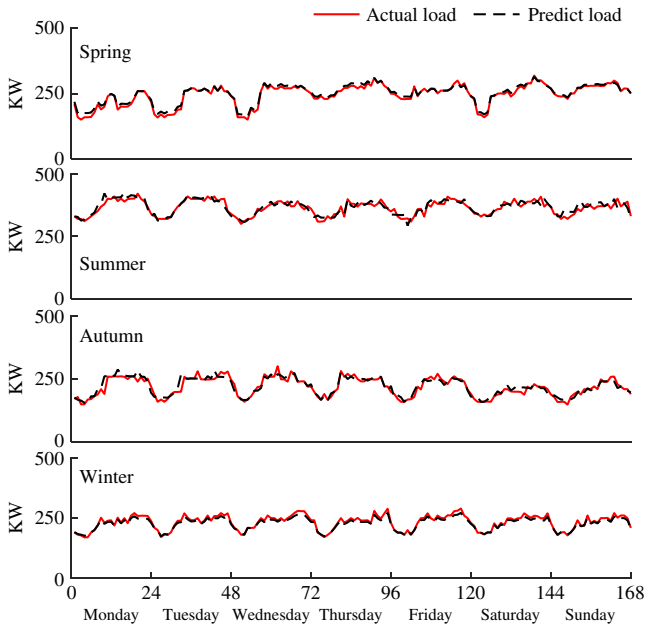


Fig. 9. Weekly load forecast at each season in a hotel

weekend. For a hotel building, the load fluctuation is gentle in day and night time, the load consumption curve is similar between weekdays and weekend.

The weekly forecasting error RMSE and MAPE of each season are presented in Table II. The forecasting MAPE for the office building is 6.67% on spring days, 2.52% on summer days, 3.82% on autumn days, and 5.23% on winter days. The biggest forecasting error is during daily peak load period in spring. For a shopping mall building load, the forecasting error is biggest in summer days; its MAPE is 7.95%, while the corresponding values are 4.62% in spring days, 2.42% in autumn days, and 2.31% in winter days. For a hotel building, the forecasting MAPE is highest in autumn days, 4.98%, and 2.78% in spring days, 3.03% in summer days and 2.54% in winter days. Different types of building load have different forecasting result according to each distinct load consumption mode. Generally speaking, the forecasting model can capture each load characteristics, and perform a high forecasting accuracy.

Table II. Weekly forecast error of different buildings

Building error	Office		Mall		Hotel	
	RMSE (KW)	MAPE (%)	RMSE (KW)	MAPE (%)	RMSE (KW)	MAPE (%)
Spring	4.42	6.67	86.31	4.62	7.41	2.78
Summer	2.43	2.52	137.82	7.95	13.78	3.03
Autumn	2.46	3.82	34.38	2.42	14.84	4.98
Winter	5.21	5.23	39.14	2.31	7.56	2.54
Average	3.63	4.56	74.41	4.32	10.90	3.33

MAPE, mean absolute percentage error; RMSE, root mean square error.

Table III. Average MAPE (%) compared with other methods

Scenario	Office	Mall	Hotel
LSTM-1 time steps	4.58	4.47	3.65
LSTM-2 time steps	**4.56**	**4.32**	**3.33**
LSTM-3 time steps	4.65	4.98	3.28
BPNN-1 time steps	13.11	10.66	5.35
BPNN-2 time steps	13.13	10.27	5.37
BPNN-3 time steps	14.12	10.65	5.64
SD-BPNN-1 time steps	9.56	4.58	4.67
SD-BPNN-2 time steps	9.32	4.54	4.70
SD-BPNN-3 time steps	9.87	5.02	4.75
ARIMA	15.36	11.23	10.69

ARIMA, autoregressive integrated moving average; BPNN, back propagation neural network; LSTM, long short-term memory; MAPE, mean absolute percentage error.

In order to illustrate the effectiveness of variable selection and the forecasting model selection, the proposed model is compared with some other algorithms including an ARIMA model, a traditional back-forward neural network (BPNN), and a back-forward neural network with similar days input selection (SD-BP). The forecasting MAPE result is shown in Table III. It can be found that the proposed LSTM strategy has the best forecast result. The result shows that when the time steps of LSTM inputs is 2, the optimal forecast performance can be achieved. For an office building, the average forecast accuracy is 13.13% with BPNN −2 time steps, 9.32% with SD-BPNN-2 time steps and 15.36% with ARIMA model. The forecasting MAPE result with the proposed model in this paper is (13.13−4.56)/13.13 = 65%, (9.32−4.56)/9.32 = 51%, and (15.36−4.56)/15.36 = 70% higher than the other methods. For a shopping mall, the average forecasting MAPE result with the proposed model in this paper is 58%, 5%, and 62% higher than the other three methods. For a hotel building, the average forecasting MAPE result with the proposed model in this paper is 38%, 30%, and 69% higher than the other three methods. It is clearly indicated that the proposed method outperforms the other methods and shows a good forecasting performance.

5. Conclusion

In this paper, the forecast of load demand at city building load systems has been discussed. Load forecast at the building level is more complex than at a concentrated load system due to the complicated load characteristics and numerous influence factors. Thus, a novel hybrid forecasting method, including an input feature selection and a LSTM-RNN forecasting model has been introduced to improve it. The impact factors are mined deeply by considering human body comfortableness of urban buildings, except normal climate features, some other features are also considered, such as index (HI), air quality, etc. Then the best candidate input features for different types of urban building

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are selected by a MRMR criterion based on Pearson's correlation coefficient. At last, a LSTM-RNN is presented for building load forecast. The result showed that appropriated input selection may help to reduce the disturbance of irrelevant variables and improve the forecasting accuracy. The proposed hybrid forecasting strategy was tested on three different types of city buildings load system, and the calculated results were compared with some other forecasted methods for 4 weeks at each season. It was illustrated that the introduced method would present a good performance with appropriate input selection and great predicting accuracy.

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Gaiping Sun (Non-member) received the B.S. and M.S. degrees from the School of Electrical and Electronic Engineering, Huazhong University of Science and Technology (HUST), in 2006 and 2008, respectively. She is currently pursuing the Ph.D. degree at ShangHai JiaoTong University. Her research interests include power generator and load forecast and electricity market.



Chuanwen Jiang (Non-member) received the M.S. and Ph.D. degrees from Huazhong University of Science and Technology, Wuhan, China, in 1996 and 2000, respectively. He completed his postdoctoral research in the Department of Electrical Engineering, Shanghai Jiaotong University, Shanghai, China. He is a Professor in the Department of Electrical Engineering, Shanghai Jiaotong University. He is now engaged in the research of reservoir dispatch, power system analysis, electricity markets, and power system economics and optimization.



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Xu Wang (Non-member) received the B.S. degree in electrical engineering from Southeast University, China, in 2010, and the Ph.D. degree from Shanghai Jiao Tong University, China, in 2016. He is currently an Assistant Professor in Shanghai Jiao Tong University, China, and also a Visiting Faculty in the Robert W. Galvin Center for Electricity Innovation at Illinois Institute of Technology (IIT), Chicago, USA. His research interests include electricity market, power flow calculation, and power system economics and optimization.



Xiu Yang (Non-member) received the Ph.D. degree in electrical engineering from Shanghai Jiao Tong University, China, in 2005. He is currently a Professor in Shanghai Jiao Tong University, China. He is a Professor with the Department of Electrical Power Engineering, Shanghai University of Electric Power. He is now engaged in the research of power system analysis, power system economics, and optimization.

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